## 7.2.6 Symmetric and Skew-symmetric Matrices

**Definition 7.17**

A square matrix $A$ is said to be **symmetric** if $A^T = A$.

That is, $A=[a_{ij}]_{n\times n}$ is a symmetric matrix, then $a_{ij} = a_{ji}$ for all $i$ and $j$.

For instance, $A=\begin{bmatrix} 3 & -6 & 9 \\ -6 & 8 & 5 \\ 9 & 5 & 2 \end{bmatrix}$ is a symmetric matrix since $A^T = A$.

Observe that transpose of $A^T$ is the matrix $A$ itself. That is $\left(A^T\right)^T = A$.

**Definition 7.18**

A square matrix $A$ is said to be **skew-symmetric** if $A^T = -A$.

If $A=[a_{ij}]_{n\times n}$ is a skew-symmetric matrix, then $a_{ij} = -a_{ji}$ for all $i$ and $j$.

Now, if we put $i = j$, then $2a_{ii} = 0$ or $a_{ii} = 0$ for all $i$. This means that all the diagonal elements of a skew-symmetric matrix are zero.

For instance, $A=\begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 4 \\ -3 & -4 & 0 \end{bmatrix}$ is a skew-symmetric matrix since $A^T = -A$ .

It is interesting to note that any square matrix can be written as the sum of symmetric and skew-symmetric matrices.

**Theorem 7.1**

For any square matrix $A$ with real number entries, $A + A^T$ is a symmetric matrix and $A - A^T$ is a skew-symmetric matrix.

**Proof**

Let $B = A + A^T$.

$$B^T = (A+A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T = B.$$

This implies $A + A^T$ is a symmetric matrix.

Next, we let $C = A - A^T$ . Then we see that

$$C^T = (A+(-A^T))^T = A^T + (-A^T)^T = A^T - (A^T)^T = A^T - A = -(A - A^T) = -C$$

This implies $A - A^T$ is a skew-symmetric matrix.

**Theorem 7.2**

Any square matrix can be expressed as the sum of a symmetric matrix and a skew-symmetric matrix.

**Proof**

Let $A$ be a square matrix. Then, we can write

$$A=\frac{1}{2}(A+A^T)+\frac{1}{2}(A-A^T).$$

From Theorem 7.1, it follows that $(A + A^T)$ and $(A - A^T)$ are symmetric and skew-symmetric matrices respectively. Since $(kA)^T = kA^T$, it follows that $\frac{1}{2}(A + A^T)$ and $\frac{1}{2}(A - A^T)$ are symmetric and skew-symmetric matrices, respectively. Now, the desired result follows.



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